EXERCISE SET 4.1

1. Using the identity

$$(a+b)^2 = a^2 + 2ab + b^2,$$

expand the following:

(i) $(7x+4y)^2$

(ii) $\left(\frac{7}{5}x + \frac{3}{2}y\right)^2$

(iii) $(2.5p + 1.5q)^2$

(iv) $\left(\frac{3}{4}s + 8t\right)^2$

(v) $\left(x + \frac{1}{2y}\right)^2$

(vi) $\left(\frac{1}{x} + \frac{1}{y}\right)^2$

Solution

(i)

$$(7x+4y)^2$$

Using

$$(a+b)^2 = a^2 + 2ab + b^2,$$

we get

$$(7x+4y)^2 = (7x)^2 + 2(7x)(4y) + (4y)^2$$

$$= 49x^2 + 56xy + 16y^2.$$

$$\boxed{49x^2 + 56xy + 16y^2}$$

(ii)

$$\left(\frac{7}{5}x + \frac{3}{2}y\right)^2$$

$$= \left(\frac{7}{5}x\right)^2 + 2\left(\frac{7}{5}x\right)\left(\frac{3}{2}y\right) + \left(\frac{3}{2}y\right)^2$$

$$= \frac{49}{25}x^2 + \frac{21}{5}xy + \frac{9}{4}y^2.$$

$$\boxed{\frac{49}{25}x^2 + \frac{21}{5}xy + \frac{9}{4}y^2}$$

(iii)

$$(2.5p + 1.5q)^2$$

$$= (2.5p)^2 + 2(2.5p)(1.5q) + (1.5q)^2$$

$$= 6.25p^2 + 7.5pq + 2.25q^2.$$

$$\boxed{6.25p^2 + 7.5pq + 2.25q^2}$$

(iv)

$$\left(\frac{3}{4}s + 8t\right)^2$$

$$= \left(\frac{3}{4}s\right)^2 + 2\left(\frac{3}{4}s\right)(8t) + (8t)^2$$

$$= \frac{9}{16}s^2 + 12st + 64t^2.$$

$$\boxed{\frac{9}{16}s^2 + 12st + 64t^2}$$

(v)

$$\left(x + \frac{1}{2y}\right)^2$$

$$= x^2 + 2x\left(\frac{1}{2y}\right) + \left(\frac{1}{2y}\right)^2$$

$$= x^2 + \frac{x}{y} + \frac{1}{4y^2}.$$

$$\boxed{x^2 + \frac{x}{y} + \frac{1}{4y^2}}$$

(vi)

$$\left(\frac{1}{x} + \frac{1}{y}\right)^2$$

$$= \left(\frac{1}{x}\right)^2 + 2\left(\frac{1}{x}\right)\left(\frac{1}{y}\right) + \left(\frac{1}{y}\right)^2$$

$$= \frac{1}{x^2} + \frac{2}{xy} + \frac{1}{y^2}.$$

$$\boxed{\frac{1}{x^2} + \frac{2}{xy} + \frac{1}{y^2}}$$

2. Using the same identity, find the values of the following:

(i) $(64)^2$

(ii) $(105)^2$

(iii) $(205)^2$

Solution

(i)

$$64^2 = (60+4)^2$$

Using

$$(a+b)^2 = a^2 + 2ab + b^2,$$

$$64^2 = 60^2 + 2(60)(4) + 4^2$$

$$= 3600 + 480 + 16$$

$$\boxed{64^2 = 4096}$$

(ii)

$$105^2 = (100+5)^2$$

$$= 100^2 + 2(100)(5) + 5^2$$

$$= 10000 + 1000 + 25$$

$$\boxed{105^2 = 11025}$$

(iii)

$$205^2 = (200+5)^2$$

$$= 200^2 + 2(200)(5) + 5^2$$

$$= 40000 + 2000 + 25$$

$$\boxed{205^2 = 42025}$$

IMPORTANT IDENTITY

$$\boxed{(a+b)^2 = a^2 + 2ab + b^2}$$